

Why PCA Sort > Value Sort: Mathematical Intuition

Value Sort: $\text{sort by mean}(\text{row}) = \sum_i x_i / D$
→ weights all dimensions equally
→ two rows with same mean but different patterns are adjacent
e.g., $[0, 1, 0, 1]$ and $[0.5, 0.5, 0.5, 0.5]$ have same mean
→ block average loses the pattern difference

PCA Sort: $\text{sort by PC1 score} = \sum_i w_i x_i$ (learned weights w_i from data)
→ weights dimensions by variance explained
→ high-variance dimensions get more weight
→ adjacent rows are similar along the most informative direction
→ block average preserves more useful information

Key insight: PCA sort minimizes within-block variance along the direction that matters most for reconstruction accuracy.

This is equivalent to 1D k-means clustering along PC1, which minimizes the quantization error of the block-mean approximation in the optimal projection direction.

Cost: $O(N)$ PCA fit (one pass) + $O(N \log N)$ sort
~4 seconds for 33M rows on CPU – done once at deployment